

particularly routing infrastructure. The principle on which mobile IP works is based on three steps: data transfer, getting recognition (discovery) and registration.

Data transfer phase is made of encapsulation and tunnelling. It works as follows.

Packet from static node to MN: Any node or host (A) can send packet to MN (B), but it has to do so with the home address of MN. IP datagram will carry IP address of A as source and home address of B as destination address.

Incoming packet is intercepted by the home agent in HN. If MN is on home network, home agent delivers the packet locally. If MN is on FN, home agent forwards the packet to the foreign agent by the process of encapsulation and tunnelling. Home agent encapsulates the whole incoming packet into a new IP packet with B's COA as destination address and then transmits the encapsulated packet as one IP datagram. The method of using an outer different destination IP address is known as tunnelling.

Foreign agent, on receiving the encapsulated packet, strips off the outer IP header. This is called decapsulation. It then encapsulates the decapsulated original IP datagram in the network-level protocol data unit (PDU), for example logical link control (LLC) frame of LAN, and delivers the packet to A on FN.

Packets from MN to any other static node: Packets from MN (B) to any other node or host (A) are sent via home agent when MN is in HN.

Packets from MN to any other node or host are sent via foreign agent when MN is in FN.

A mix of techniques described above is used for the exchange of packets between two different MNs.

The principle, as stated above, will work only when MN is available to register its mobility with the foreign agent in FN. It has to register when it enters FN as well as when it leaves FN. This is done by a process called getting

recognition (discovery) and registration.

Getting recognition is a process by which MN registers its location with the appropriate agent, either home or foreign. MN finds a mobility agent by two methods: agent advertisement and agent solicitation message.

Agent advertisement is a process by which foreign and home agents exchange their respective presence periodically using agent advertisement message. MN in a subnet then receives agent advertisements from either its home or foreign agent. This is one way for MN to discover its location.

In the second method, MN may use agent solicitation message for a quick reply for locating the agent. In either case, this process allows MN to identify its current mobility agents and then obtain one or more COAs.

Using this method, it also learns about any special services provided by a foreign agent and determines whether it is connected to its HN or FN.

Registration is a process by which having received COA, MN has to register with home agent. As MN roams, it is connected to a series of one or more FNs. The main purpose of this registration is to inform home agent of the current location of MN for correct forwarding of packets. This process provides a link between home and COAs, and defines the lifetime of registration.

This is done by sending binding updates from MN to home agent. For this three values, namely, MN home static IP address, COA and lifetime are used. Home agent sends a registration reply to foreign agent, who forwards this reply to MN. This operation is shown in Figure 5. Encryption is used to authenticate the registration information.

A key concern with registration is security. The default authentication algorithm uses key-based MD5 (message digest version 5) to produce a 128-bit message digest.

When MN is in FN and is registered with home agent, home agent will work as a proxy for MN. When HN receives packets addressed to MN, it intercepts those packets (using Proxy ARP) and encapsulates them with overhead COA.

Encapsulated packet is then sent to COA, which is the current IP address the foreign agent has registered with MN. Foreign agent will decapsulate the packet and forward it to MN, because it knows exactly where MN is. Encapsulation is the method used by home agent to deliver information to MN by putting an extra IP header on top of the packet.

Tunnelling is a process by which the encapsulated packet is sent to MN when it is in FN (Figure 5). Two forms of encapsulation are specified in related standards request.

Getting recognition (discovery) makes use of ICMP. This is done by appending appropriate extension to ICMP headers. ICMP is a connection-less protocol that suits this operation. Registration requires a protocol that communicates between an application on MN and that is the home agent. Thus, it requires a transport-level protocol. As it is a simple request-and-response transaction, overhead of connection-oriented TCP is not required. Therefore, UDP is used as the transport protocol for registration. Tunnelling is done at IP level. 

This article is dedicated by the author to Prof. A.K. Chowdhury, who is known as the father of computer science in India.

 By: Prof. Chandan Tilak Bhunia

The author is PhD in computer engineering from Jadavpur University. He is also fellow of Institution of Electronics and Telecommunication Engineers, fellow of Institution of Engineers (India) and fellow of Computer Society of India.

The article was originally published in the August 2019 issue of Electronics For You.